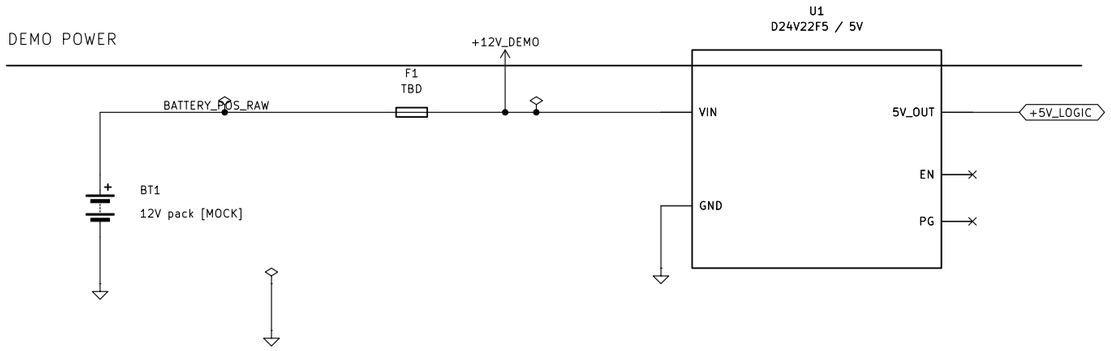
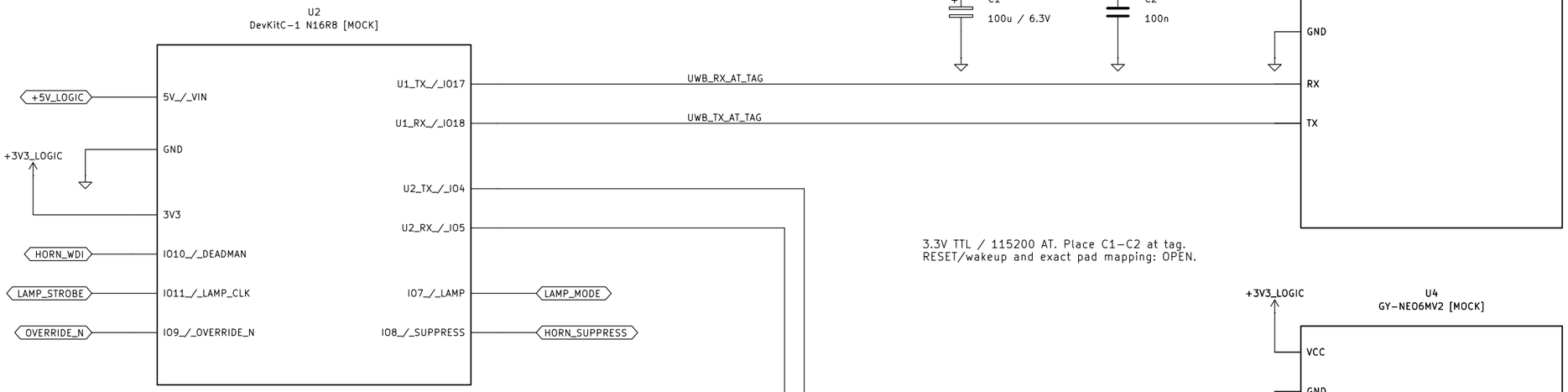
VEHICLE MODULE / POWER AND CONTROLLER

DEMO POWER

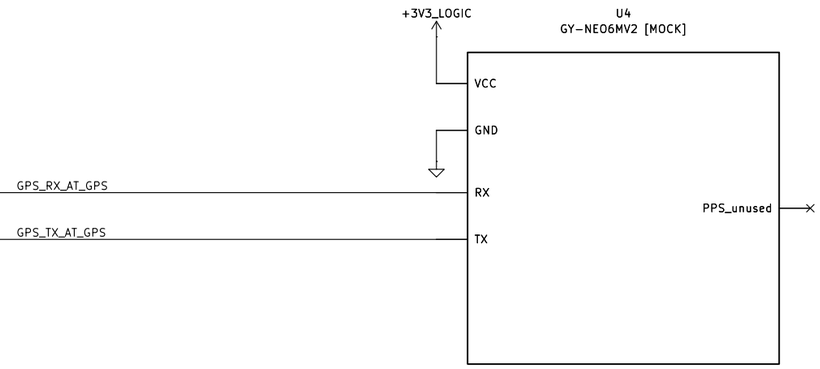


Dedicated 12V demo pack; F1 selected after load test.
Buck input: 12V nominal. Output: 5V –> DevKit VIN.
Onboard LDO supplies the separate 3.3V logic rail.
Disconnect USB power when VIN is powered.

No 5V on GPIO. Never apply >3.6V to 3V3.
Battery limits, fuse rating and regulator loading: OPEN.



3.3V TTL / 115200 AT. Place C1–C2 at tag.
RESET/wakeup and exact pad mapping: OPEN.



CONTROLLER ASSIGNMENTS

UART0 IO43 / IO44: programming and debug only.
UART1 IO17 TX / IO18 RX: tag.
UART2 IO4 TX / IO5 RX: GPS; explicit matrix mapping.
IO7 mode data; IO11 lamp strobe; IO8 suppress;
IO10 deadman pulses; IO9 override sense.
All selected IO are non–strapping; no ADC2 analogue use.
Keep 0/3/45/46 and memory 35–37 free.

GPS: 9600 NMEA; broad context only, never sole zone input.
2.5m CEP / up to 5Hz are datasheet values.
Check breakout dropout: chip rail must remain 2.7–3.6V.
[MOCK] = functional interface; confirm actual board/pad revision.

Demo outputs and override

File: Demo Outputs and Override.kicad_sch

Lamp mode and manual selection

File: Lamp Mode and Manual Selection.kicad_sch

DEMO ONLY | Interfaces and open items: project notes
Power, controller and ranging interfaces
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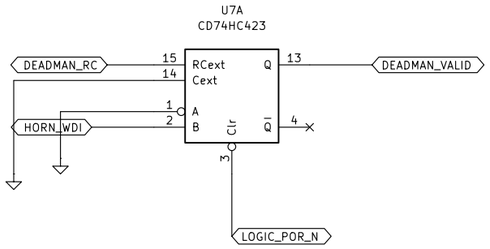
Sheet: /
File: Vehicle Module Electronic Schematic.kicad_sch

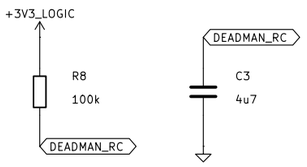
Title: Vehicle Module Electronic Schematic

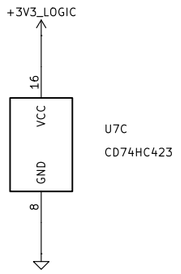
Size: A3	Date: 2026–10–24	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/3

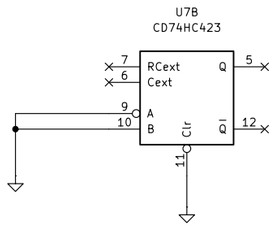
VEHICLE MODULE / HORN CUT-OFF AND OVERRIDE

EXTERNAL DEADMAN / 3.3V



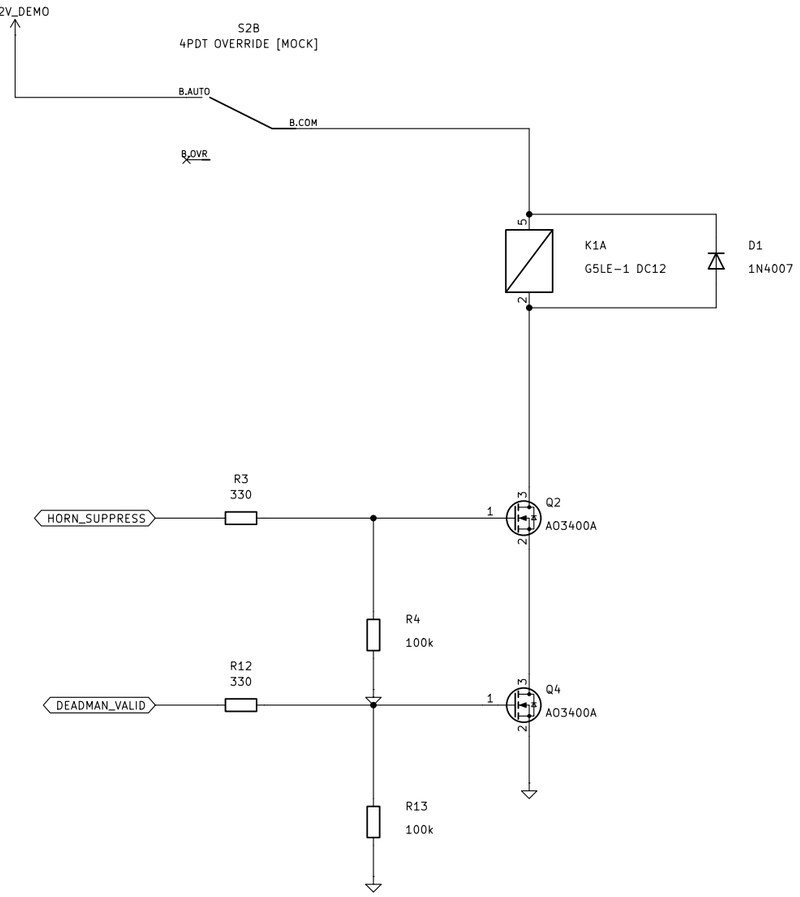






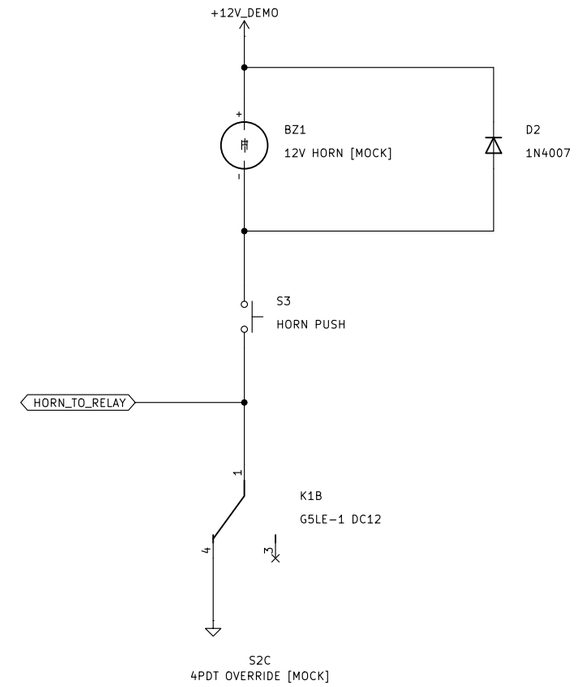
IO10 pulses only while a validated command is fresh.
A stuck HIGH or LOW has no retrigger edge. U7 Q then falls.
Nominal RC timing requires 3.3V/tolerance measurement.
U6 reset holds the deadman off at power-up.

RELAY DRIVER / FLYBACK



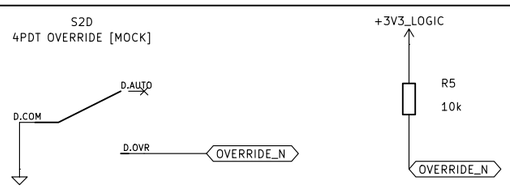
Q2 command AND Q4 deadman must conduct for coil energy.
S2B disconnects coil supply in OVERRIDE; flyback D1 is local.

DRIVER HORN / NC CONTACT AND BYPASS



S2C bypasses K1 NC during override, including a welded
open contact. Switch contact rating vs horn inrush is OPEN.

OVERRIDE SENSE AND LIMITS



UNKNOWN, PREDICTED, stale or link loss: no new lamp clock;
stop IO10 pulses so coil releases. Reject expired or replayed data.
OVERRIDE: manual lamp choice, coil cut, horn bypass, sense LOW.
All S2 sections move together and must break before make.
Demo hardware only. No connection to vehicle horn/headlamp.

DEMO ONLY | Interfaces and open items: project notes
Horn cutoff and physical override
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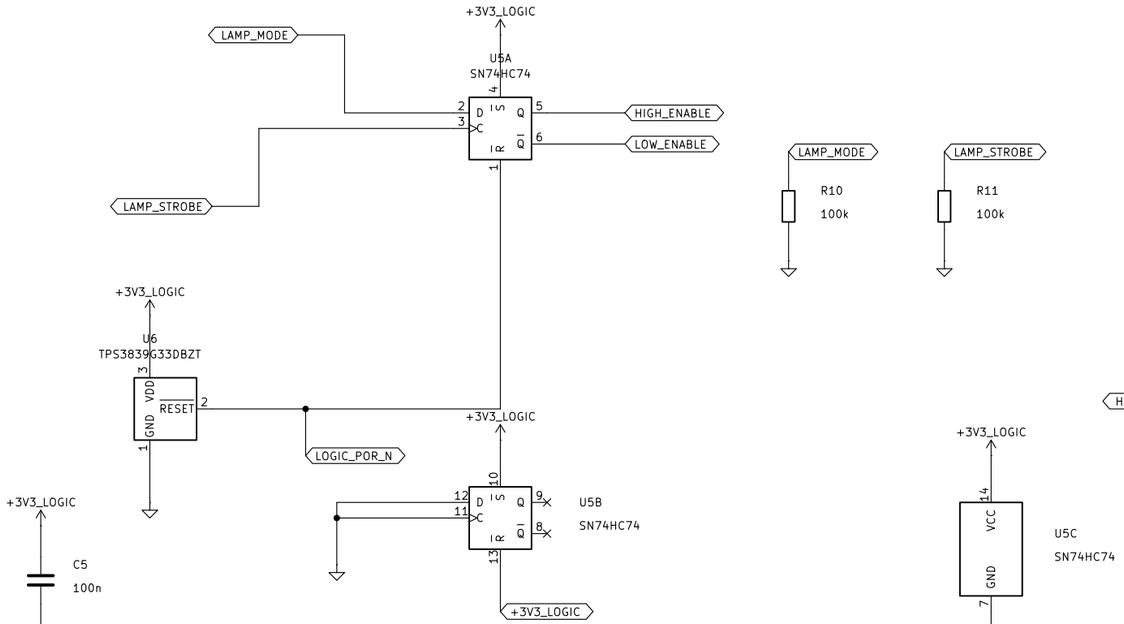
Sheet: /Demo outputs and override/
File: Demo Outputs and Override.kicad_sch

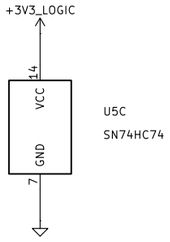
Title: Vehicle Module Electronic Schematic

Size: A3	Date: 2026-10-24	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 2/3

VEHICLE MODULE / STORED LAMP MODE AND MANUAL SELECTION

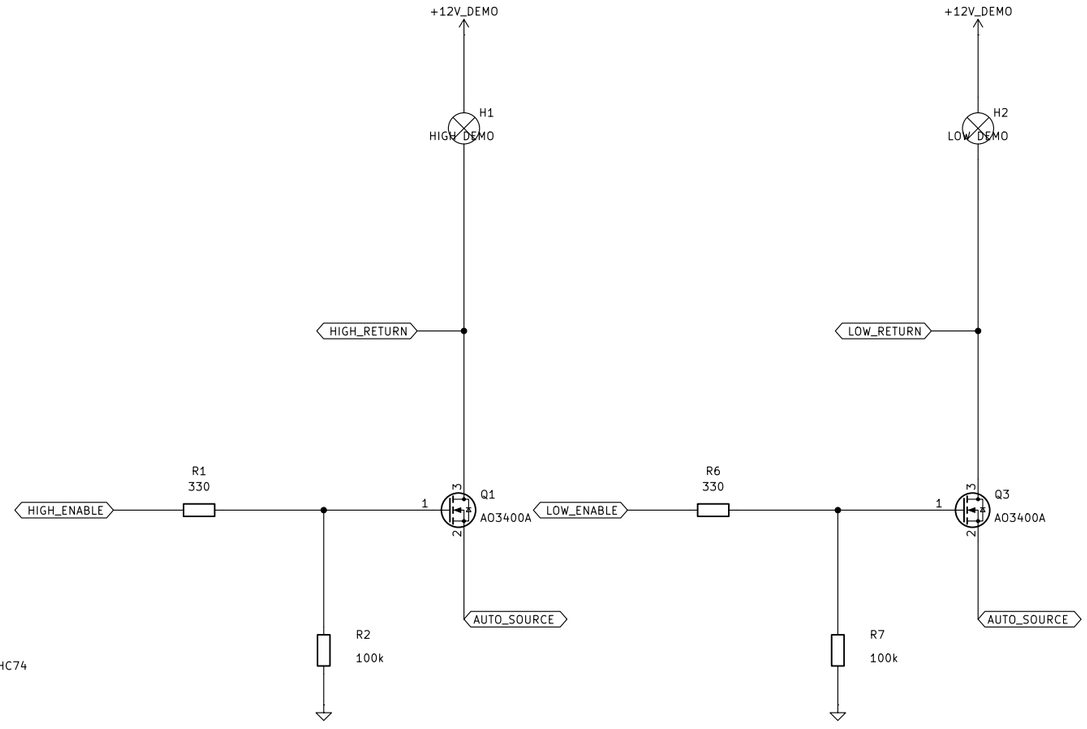
MODE STORAGE / 3.3V





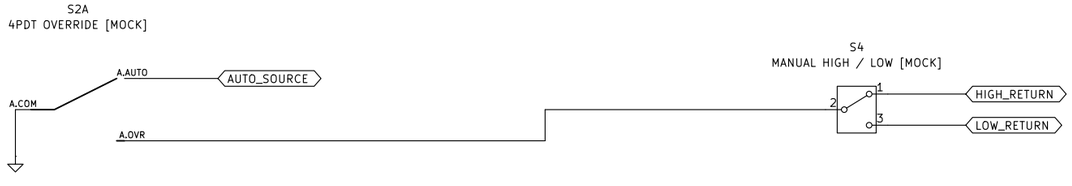
U6 releases clear after logic rail recovery. Power-up selects LOW.
ESP reset with the 3.3V rail alive does not clear U5.
Stale/UNKNOWN input: no clock edge; U5 retains last mode.
Loss of logic power clears the stored mode; no lamp hold is claimed then.

INDEPENDENT HIGH / LOW DEMO LAMPS



Q and /Q select one steady mode. Check transition overlap
and lamp current/inrush with actual demo loads.

BREAK-BEFORE-MAKE DRIVER OVERRIDE



AUTO: S2A grounds both driver sources. OVERRIDE: S2A opens both
MOSFET source returns and grounds S4 common. S4 selects HIGH or LOW.
S2/S4 terminal maps and DC current ratings: OPEN until selected parts.

DEMO ONLY | Interfaces and open items: project notes
Stored lamp mode and manual selection
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Sheet: /Lamp mode and manual selection/
File: Lamp Mode and Manual Selection.kicad_sch

Title: Vehicle Module Electronic Schematic

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